

LS-LSTER :

```
vboxuser@Ubuntu:~/link$ ls
vboxuser@Ubuntu:~/link$ vi original.txt
vboxuser@Ubuntu:~/link$ echo "hi everyone and it's practise of hard link and soft link "
hi everyone and it's practise of hard link and soft link
vboxuser@Ubuntu:~/link$ ls -lstr
total 0
0 -rw-rw-r-- 1 vboxuser vboxuser 0 May 17 12:35 original.txt
```

HARD LINK :

```
original.txt
vboxuser@Ubuntu:~/link$ ln original.txt hard_link.txt
vboxuser@Ubuntu:~/link$ ls
hard_link.txt  original.txt
vboxuser@Ubuntu:~/link$
```

LS-LSTR:

```
vboxuser@Ubuntu:~/link$ ln original.txt team2_data.txt
vboxuser@Ubuntu:~/link$ ls -lstr
total 0
0 -rw-rw-r-- 3 vboxuser vboxuser 0 May 17 12:35 original.txt
0 -rw-rw-r-- 3 vboxuser vboxuser 0 May 17 12:35 hard_link.txt
0 -rw-rw-r-- 3 vboxuser vboxuser 0 May 17 12:35 team2_data.txt
vboxuser@Ubuntu:~/link$ cat original.txt
vboxuser@Ubuntu:~/link$
```

SOFT LINK :

```
vboxuser@Ubuntu:~/link$ cat orriginal.txt
soft link
vboxuser@Ubuntu:~/link$ ln -s original.txt new_data.txt
vboxuser@Ubuntu:~/link$ ls -lstr
total 4
0 -rw-rw-r-- 3 vboxuser vboxuser 0 May 17 12:35 original.txt
0 -rw-rw-r-- 3 vboxuser vboxuser 0 May 17 12:35 hard_link.txt
0 -rw-rw-r-- 3 vboxuser vboxuser 0 May 17 12:35 team2_data.txt
4 -rw-rw-r-- 1 vboxuser vboxuser 10 May 17 12:46 orriginal.txt
0 lrwxrwxrwx 1 vboxuser vboxuser 12 May 17 12:48 new_data.txt -> original.txt
vboxuser@Ubuntu:~/link$
```

NOHUP AND TOP :

```
top - 12:54:35 up 6:00, 1 user, load average: 0.01, 0.02, 0.00
Tasks: 217 total, 1 running, 216 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.6 us, 2.4 sy, 0.0 ni, 95.8 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 3398.8 total, 498.9 free, 1510.4 used, 1645.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1888.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3039	vboxuser	20	0	4040200	422576	172888	S	19.6	12.1	1:38.49	gnome-s+
3161	vboxuser	20	0	470536	11784	7844	S	0.7	0.3	0:04.66	ibus-da+
4546	vboxuser	20	0	1816000	393176	137812	S	0.7	11.3	0:33.28	ptyxis
6075	vboxuser	20	0	1706368	266940	146980	S	0.3	7.7	0:04.61	nautilus
6786	root	20	0	0	0	0	I	0.3	0.0	0:00.69	kworker+
7114	vboxuser	20	0	22064	6372	4244	R	0.3	0.2	0:00.01	top
1	root	20	0	25800	17252	11724	S	0.0	0.5	0:05.94	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.06	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_wo+
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker+
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker+
14	root	20	0	0	0	0	S	0.0	0.0	0:00.13	ksoftir+

WILL SHOW THE BACKGROUND PROCESS RUNNING

PS:

```
vboxuser@Ubuntu:~$ ps
  PID TTY          TIME CMD
  4588 pts/0    00:00:00 bash
  7119 pts/0    00:00:00 ps
vboxuser@Ubuntu:~$
```

PS – AUX :

```

7119 pts/0    00:00:00 ps
vboxuser@ubuntu:~$ ps -aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.4 25800 17252 ?        Ss   06:53   0:05 /usr/lib/syst
root           2  0.0  0.0      0     0 ?        S    06:53   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    06:53   0:00 [pool_workque
root           4  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-rc
root           5  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-sy
root           6  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-kv
root           7  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-sl
root           8  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-ne
root          12  0.0  0.0      0     0 ?        I    06:53   0:00 [kworker/u8:0
root          13  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-mm
root          14  0.0  0.0      0     0 ?        S    06:53   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    06:53   0:02 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    06:53   0:00 [rcu_exp_par_
root          17  0.0  0.0      0     0 ?        S    06:53   0:00 [rcu_exp_gp_k
root          18  0.0  0.0      0     0 ?        S    06:53   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    06:53   0:00 [kprobe-optim
root          20  0.0  0.0      0     0 ?        S    06:53   0:00 [idle_inject/
root          21  0.0  0.0      0     0 ?        S    06:53   0:00 [cpuhp/0]
root          22  0.0  0.0      0     0 ?        S    06:53   0:00 [cpuhp/1]
root          23  0.0  0.0      0     0 ?        S    06:53   0:00 [idle_inject/
root          24  0.0  0.0      0     0 ?        S    06:53   0:00 [migration/1]
root          25  0.0  0.0      0     0 ?        S    06:53   0:00 [ksoftirqd/1]
root          30  0.0  0.0      0     0 ?        S    06:53   0:00 [kdevtmpfs]
root          31  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-in
root          32  0.0  0.0      0     0 ?        I    06:53   0:00 [rcu_tasks_kt
root          33  0.0  0.0      0     0 ?        I    06:53   0:00 [rcu_tasks_ru
root          34  0.0  0.0      0     0 ?        S    06:53   0:00 [kauditd]
root          35  0.0  0.0      0     0 ?        S    06:53   0:00 [khungtaskd]
root          36  0.0  0.0      0     0 ?        S    06:53   0:00 [oom_reaper]
root          38  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-wr
root          39  0.0  0.0      0     0 ?        S    06:53   0:02 [kcompactd0]
root          40  0.0  0.0      0     0 ?        SN   06:53   0:00 [ksmd]
root          42  0.0  0.0      0     0 ?        SN   06:53   0:00 [khugepaged]
root          43  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-kb
root          44  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-bl
root          45  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-ki
root          46  0.0  0.0      0     0 ?        S    06:53   0:00 [irq/9-acpi]
root          47  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-tp
root          48  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-at
root          49  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-md
root          50  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-md
root          51  0.0  0.0      0     0 ?        I<   06:53   0:00 [kworker/R-md

```

KILL:

```

root          1538  0.0  0.0      0     0 ?        Ss   06:14   0:00 /init
ubuntu        1538  0.0  0.0      0     0 pts/2    Ss   06:14   0:02 -bash
root          60879  0.0  0.2 293008 20400 ?        Ssl  10:23   0:00 /usr/libexec/packagekitd
root          60883  0.0  0.0 234512 6632 ?        Ssl  10:23   0:00 /usr/libexec/polkitd --no-debug
root          64491  0.0  0.1 15436 8700 ?        Ss   10:26   0:00 sshd: /usr/sbin/sshd -D [listener]
ubuntu        155745  0.0  0.0 7804 3760 pts/2    T    16:43   0:00 top
ubuntu        165966  0.0  0.0 13548 7660 pts/2    S    17:25   0:00 python3 data_processing.py
ubuntu        166049  0.0  0.0 7804 3588 pts/2    T    17:26   0:00 top
ubuntu        167117  0.0  0.0 7756 3524 pts/2    R+   17:30   0:00 ps -aux
ubuntu@gowtham:~$ kill -9 165966
ubuntu@gowtham:~$
[2] Killed
ubuntu@gowtham:~$ nohup python3 data_processing.py >> data_process.log
ubuntu@gowtham:~$

```

PS -AUX |GREP FILE NAME :

```
Try 'grep --help' for more information.
vboxuser@Ubuntu:~$ ps -aux | grep data.log
vboxuser    7183  0.0  0.0 18132 2676 pts/0    S+   13:04   0:00 grep --color=auto data.log
vboxuser@Ubuntu:~$
```

DOWNLOAD FILE :

```
ubuntu@gowtham: $
ubuntu@gowtham: $ wget "https://wonderfulengineering.com/wp-content/uploads/2014/06/Linux-wallpapers-13.jpg"
--2025-06-09 17:34:01-- https://wonderfulengineering.com/wp-content/uploads/2014/06/Linux-wallpapers-13.jpg
Resolving wonderfulengineering.com (wonderfulengineering.com)... 209.59.130.145
Connecting to wonderfulengineering.com (wonderfulengineering.com)|209.59.130.145|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 131379 (128K) [image/jpeg]
Saving to: 'Linux-wallpapers-13.jpg'

Linux-wallpapers-13.jpg      100%[=====] 128.30K  125KB/s  in 1.0s
2025-06-09 17:34:04 (125 KB/s) - 'Linux-wallpapers-13.jpg' saved [131379/131379]

ubuntu@gowtham: $
```